Rational Expressions

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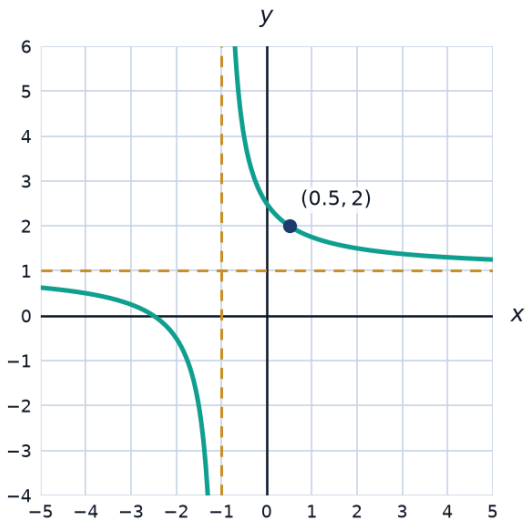
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of the equation $f(x) = 0$?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the polynomial $P(x) = x^3 - 6x^2 + kx + 12$. If $(x - 3)$ is a factor of $P(x)$, and the sum of the roots of $P(x) = 0$ is S, find the value of k. Additionally, determine the sum of the roots S given that the roots are α, β, γ.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The rational expression $\frac{2x^2 + ax - 7}{x + 4}$ is equivalent to $2x + b + \frac{5}{x + 4}$ for all $xeq -4$, where a and b are constants. What is the value of $a + b$?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function f is defined by $f(x) = (x + 3)^2 - 5$. What are the coordinates of the vertex of the graph of $y = f(x)$ in the xy-plane?

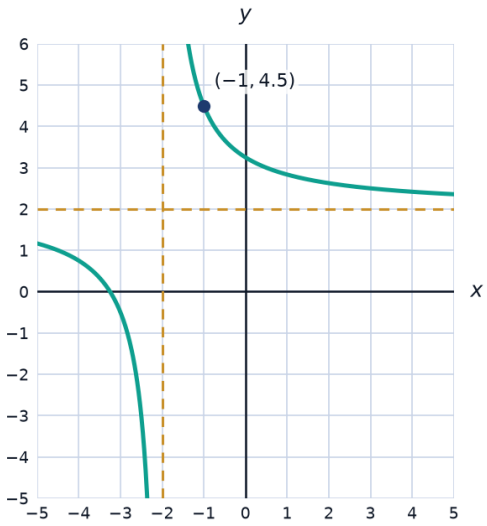
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quadratic function $f(x) = ax^2 + bx + c$ has a vertex at the point $(2, -5)$. If the graph of $f(x)$ passes through the point $(4, 3)$, find the value of a.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Factor the quadratic expression: $x^2 + 5x + 6$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the rational expression: $\frac{x^2-4}{x-2}$. If $xeq2$, what is the simplified form of this expression?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

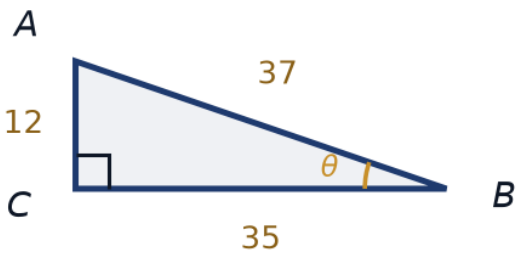
Given the polynomial $P(x) = ax^3 - 2x^2 + bx + 5$, if $P(x)$ is divisible by $(x - 1)$ and the remainder when $P(x)$ is divided by $(x + 2)$ is -9, find the value of $a + b$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If the expression $\frac{\sqrt[4]{x^3 \cdot \sqrt{x^k}}}{x^{-\frac{1}{4}}}$ is equivalent to x^2 for all $x > 0$, what is the value of the constant k?

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Triangle ABC is mapped to triangle $A'B'C'$ on the coordinate plane as shown below. Which coordinate rule represents this transformation?



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $f(x) = \sqrt{x^2 - 4x + 4}$ and $g(x) = \sqrt[3]{x^3 + 6x^2 + 12x + 8}$. If $h(x) = f(x) \cdot g(x)$, find the value of $h(-3)$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function defined by $f(x) = \sqrt{ax + b}$. If the graph of $f(x)$ passes through the points $(1, 3)$ and $(5, 5)$, determine the value of a.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the figure below, triangle ABC is a right triangle with right angle at C, and altitude CD is drawn to hypotenuse AB. The length of altitude CD is 6, $AD = x$, and $DB = x + 5$. What is the length of hypotenuse AB?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the function $y = a \cdot b^x$, where $ae0$ and $b > 0, be1$, passes through the points $(1, 6)$ and $(3, 54)$. Determine the exact value of a and the exact value of b.

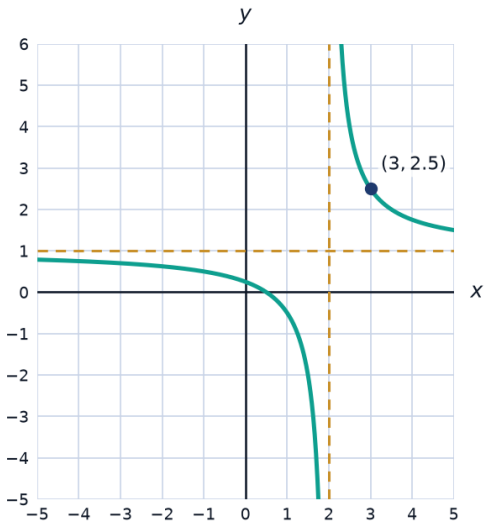
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(0, 3)$ and $(2, 75)$. Determine the value of a.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

The graph of the function $f(x) = \sqrt{ax + b}$ passes through the points $(1, 3)$ and $(5, 5)$. Find the value of a and the value of b.



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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

The graph of the exponential function $f(x) = a \cdot b^x + c$ passes through the points $(0, 3)$ and $(1, 7)$. If the horizontal asymptote of the graph is $y = 1$, find the value of a and b.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Let $P(x)$ be a polynomial such that $P(x^2 + 1) = x^4 + 3x^2 + 2$. If $P(x)$ can be expressed in the form $ax^2 + bx + c$, find the value of b.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

Simplify the complex rational expression: $\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}}$, where $xeqy$ and $xyeq0$.

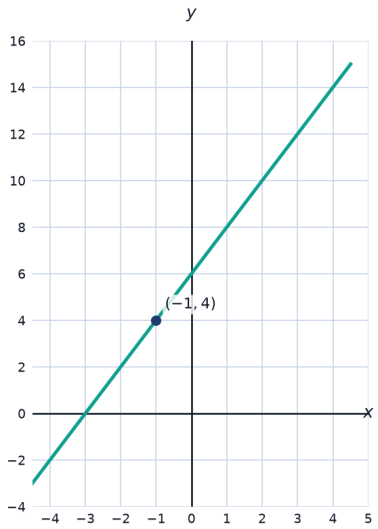
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Linear Equations & Lines

[Rational Expressions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 6$ in key features of quadratic functions, evaluate $f(2)$.



a. 10

b. 12

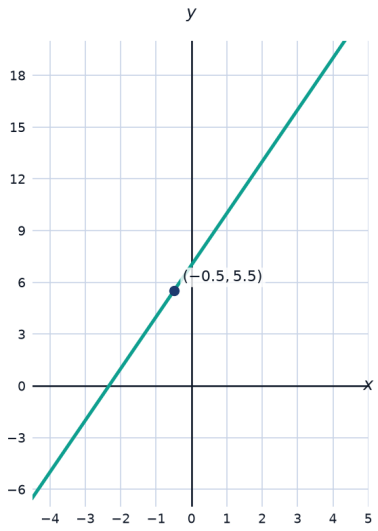
c. 4

d. 14

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in solving quadratic equations by factoring, evaluate $f(2)$.



a. 13

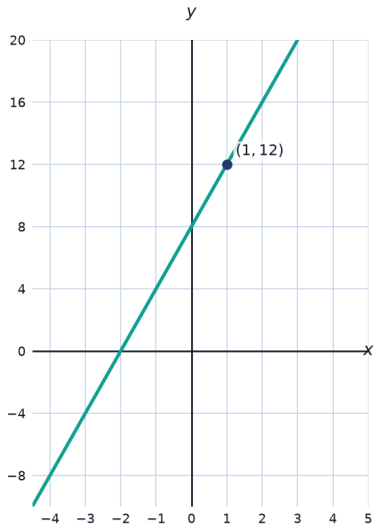
b. 16

c. 6

d. 17

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in solving quadratic equations using square roots, evaluate $f(2)$.



a. 16

b. 20

c. 8

d. 20

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for key features of quadratic functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solving quadratic equations by factoring consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for solve quadratic equations using the quadratic formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 9$ in solve quadratic equations using the quadratic formula, evaluate $f(4)$.

a. 29

b. 34

c. 20

d. 33

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for graphing quadratic functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 3$ in graphing quadratic functions, evaluate $f(4)$.

a. 27

b. 33

c. 24

d. 31

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 4$ in transformations of quadratics, evaluate $f(3)$.

a. 10

b. 12

c. 6

d. 14

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all distinct real solutions to the equation $\frac{x}{x^2+4} + \frac{x}{x^2-3x+4} = \frac{1}{2}$?

a. 6

b. 7

c. 3

d. $3+\sqrt{5}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following is equivalent to the expression $\frac{5}{x} + \frac{2}{3x}$ for all $x \neq 0$?

a. $\frac{17}{3x}$

b. $\frac{7}{4x}$

c. $\frac{7}{3x}$

d. $\frac{10}{3x^2}$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following expressions is equivalent to $\frac{x^2-4}{x-2}$ for all values of x where the expression is defined?

a. $x - 2$

b. $x + 2$

c. x

d. 2

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which rational expression is equivalent to $\frac{2x+4}{x+2}$?

a. $2x$

b. $x+2$

c. 2

d. $\frac{2}{x}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify $\frac{x^2-25}{x-5}$, then evaluate the result at $x=10$.

a. 14

b. 10

c. 17

d. 15

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What value makes $\frac{x}{x-3}$ undefined?

a. 3

b. 0

c. -3

d. 1

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following expressions is equivalent to $\frac{(x^4)^3}{x^5}$ for all $x \neq 0$?

a. x^2

b. x^7

c. $x^{\frac{12}{5}}$

d. x^{17}

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Factoring Polynomials

Which of the following expressions is equivalent to $\sqrt{25x^4}$?

a. $\sqrt{5}x^2$

b. $5x^2$

c. $5x$

d. $25x^2$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

Rationalize the denominator: $\frac{8}{\sqrt{2}}$.

a. $\frac{8\sqrt{2}}{3}$

b. $8\sqrt{2}$

c. $\frac{8\sqrt{2}}{2}$

d. $\frac{9\sqrt{2}}{2}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Linear Equations & Lines

Write $x^{2/3}$ using a radical.

a. $\sqrt{x^{3/2}}$

b. $\sqrt[2]{x^3}$

c. $(\sqrt{x})^3$

d. $\sqrt[3]{x^2}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Statistics & Data Analysis

Which is equivalent to $x^{-1/2}$?

a. $\sqrt{x}$

b. $\frac{1}{\sqrt{x}}$

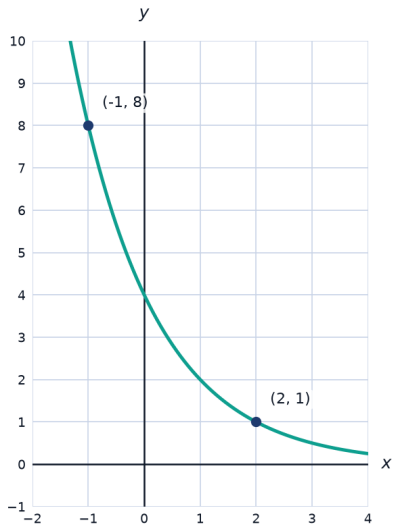
c. $\frac{1}{x^2}$

d. $-\sqrt{x}$

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42 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 2.01 Quadratic Functions**

The graph of the exponential function $f(x) = a \cdot b^x$, where a and b are positive constants, is shown in the xy-plane below. The graph passes through the points $(-1, 8)$ and $(2, 1)$. If $f(k) = \frac{1}{16}$, what is the value of k?



a. 6

b. 5

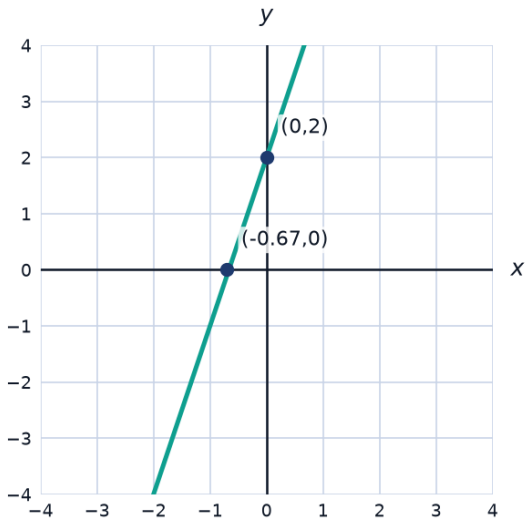
c. 4

d. -4

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Exponents & Polynomials

A line of best fit for data has equation $y = 3x + 2$. Predict y when $x = 5$.



a. 20

b. 17

c. 15

d. 13

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Functions & Transformations

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

b. 88%

c. 92%

d. 96%

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A sequence of positive real numbers $(a_n)_{n \geq 1}$ satisfies $a_1 = 1$, $a_2 = 3$, and the recurrence relation $a_{n+1} = \frac{a_n^2 + 2}{a_{n-1}}$ for all integers $n \geq 2$. What is the value of the infinite series $\sum_{n=1}^{\infty} \frac{1}{a_n a_{n+1}}$?

a. $\frac{\sqrt{3}-1}{2}$

b. $\sqrt{3}-1$

c. $\frac{3-\sqrt{3}}{2}$

d. $\frac{2-\sqrt{3}}{2}$

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